

Student Mental Health Support Platform

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A Privacy-First Digital Support System for University Students

INTRODUCTION

University students frequently experience emotional and academic stress, but institutional mental health services are not always visible, accessible, or trusted. The project survey showed a gap between student need and help-seeking behavior: students report emotional difficulties, but many do not use existing support services. Therefore, the problem is not only technological; it involves trust, anonymity, awareness, institutional coordination, and support capacity.

Problem Highlights

- 88% reported anxiety or emotional difficulties
- 68% had never sought help
- 0% students rated current services as easy to access
- 56% did not know how accessible current services were
- 64% preferred anonymous support

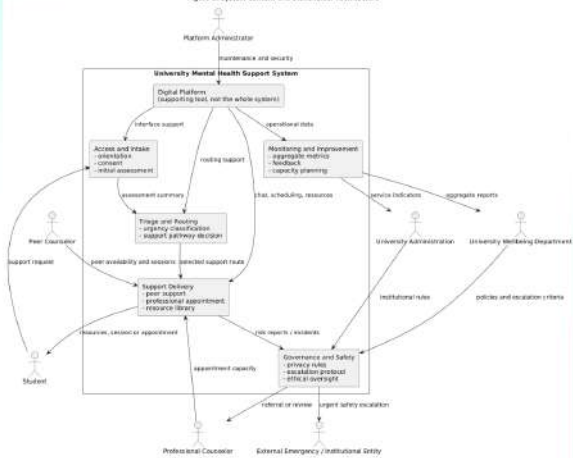
GOAL

The goal of this project is to design a digital mental health support platform that reduces barriers to access for university students. The expected product is a scalable, anonymous, and modular system that provides peer support, professional counseling access, and educational mental health resources.

PROPOSED SOLUTION

The proposed platform is designed as one component of a broader university wellbeing ecosystem. Students enter through an anonymous intake process and complete an initial needs assessment. Low-intensity cases are guided toward educational resources, moderate cases are routed to trained peer counselors, and higher-need cases are referred to professional counseling services. Privacy, consent, data minimization, role-based access, and monitoring operate as cross-cutting controls throughout the system.

Figure 1. System Context and Stakeholder Architecture



Core system logic: request, consent, assessment, triage, assignment, escalation, follow-up, and improvement.

EXPERIMENTS

The validation stage used discrete-event simulation to test whether the design remains coherent under different operational conditions. The model focused on student entry, trust-sensitive registration, peer counselor assignment, counselor workload, response time, and dropout behavior.

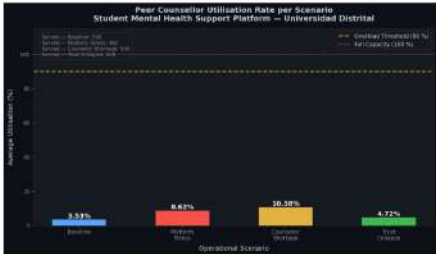
Scenario	Purpose
Baseline	Normal demand and
Midterm/Finals Stress	Academic peak demand,
Counselor Shortage	Reduced peer counselor
Trust Collapse	Lower user trust and

Simulation parameters:
500 students · 30 peer counselors · 2.5 peak multiplier · 5 min response target · 20-session burnout threshold · 0.85 baseline trust level.

RESULTS

Main finding: the Midterm/Finals Stress scenario produced 18 dropouts even when average waiting time remained near zero. This means that average wait time alone is not enough to evaluate system safety; the platform needs real-time queue monitoring and proactive messages during peak academic periods

Scenario	Utilization	Dropouts	Served
Baseline	3.53%	0	500
Midterm/Finals Stress	8.62%	18	482
Counselor Shortage	10.58%	0	500
Trust Collapse	4.72%	0	500



DISCUSSION

The simulations validate the general feasibility of the architecture under baseline conditions. However, they also show that the system depends on operational governance. Stress periods require academic-calendar-aware surge planning, counselor shortage requires capacity monitoring and workload caps, and trust collapse requires privacy-by-design, transparent communication, and incident response protocols.

REFERENCES

- World Health Organization. *Mental health and well-being*. 2022.
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CONCLUSIONS

The project designed and validated a Student Mental Health Support Platform as part of a broader university wellbeing ecosystem. The proposal is successful because it translates survey evidence into a system architecture focused on access, anonymity, multi-channel support, referral, risk management, and continuous improvement. Future work should include a controlled pilot, real operational data collection, escalation validation with the wellbeing department, and refinement of the simulation model using observed student behavior.